

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 14:

Money, Banking and Monetary policy

► What you will learn in this chapter



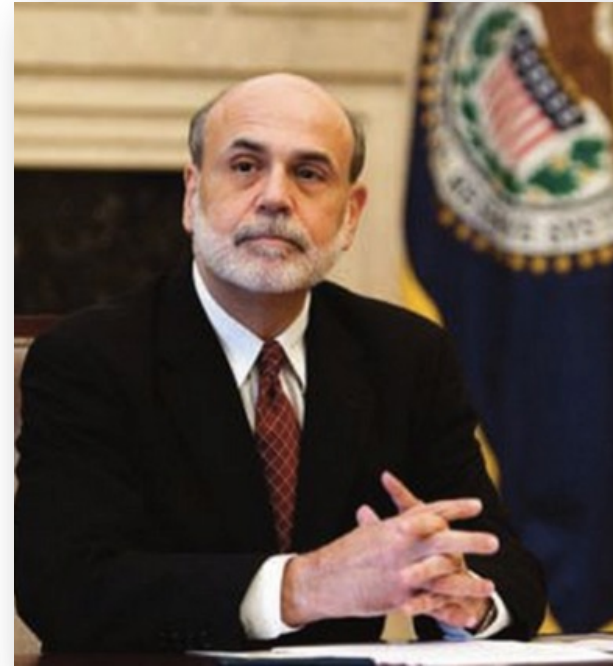
- What the money demand curve is
- Why the liquidity preference model determines the interest rate in the short run
- How the Federal Reserve implements monetary policy, moving the interest rate to affect aggregate output
- Why monetary policy is the main tool for stabilizing the economy
- How the behavior of the Federal Reserve compares to that of other central banks
- Why economists believe in monetary neutrality—that monetary policy affects only the price level, not aggregate output, in the long run

THE MOST POWERFUL PERSON IN GOVERNMENT

Janet Yellen, Current Chair



Ben Bernanke, Former Chair



The chairmanship of the Federal Reserve is arguably the most powerful position in the U.S. government

THE DEMAND FOR MONEY



There is a price to be paid for the convenience of holding money.

The Opportunity Cost of Holding Money

We all carry some cash around for the convenience.

When we do, we give up interest income we'd collect if that spending power were in an interest-bearing asset like a bond.

INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Individuals and firms trade off the benefit of holding cash versus the benefit of holding interest-bearing nonmonetary assets.

Selected interest rates, 2007 (before the financial crisis)

One-month certificates of deposit (CDs)	5.30%
Interest-bearing demand deposits	2.30%
Currency	0

Source: Federal Reserve Bank of St. Louis

INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Short-term interest rates: the interest rates on financial assets that mature within six months or less.

Long-term interest rates: interest rates on financial assets that mature a number of years in the future.

The ***higher*** the interest rate, the ***higher*** the opportunity cost of holding money.

The ***lower*** the interest rate, the ***lower*** the opportunity cost of holding money.

INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Although there are many different interest rates and they don't move exactly together, as a rule **most short-term rates tend to move in the same direction.**

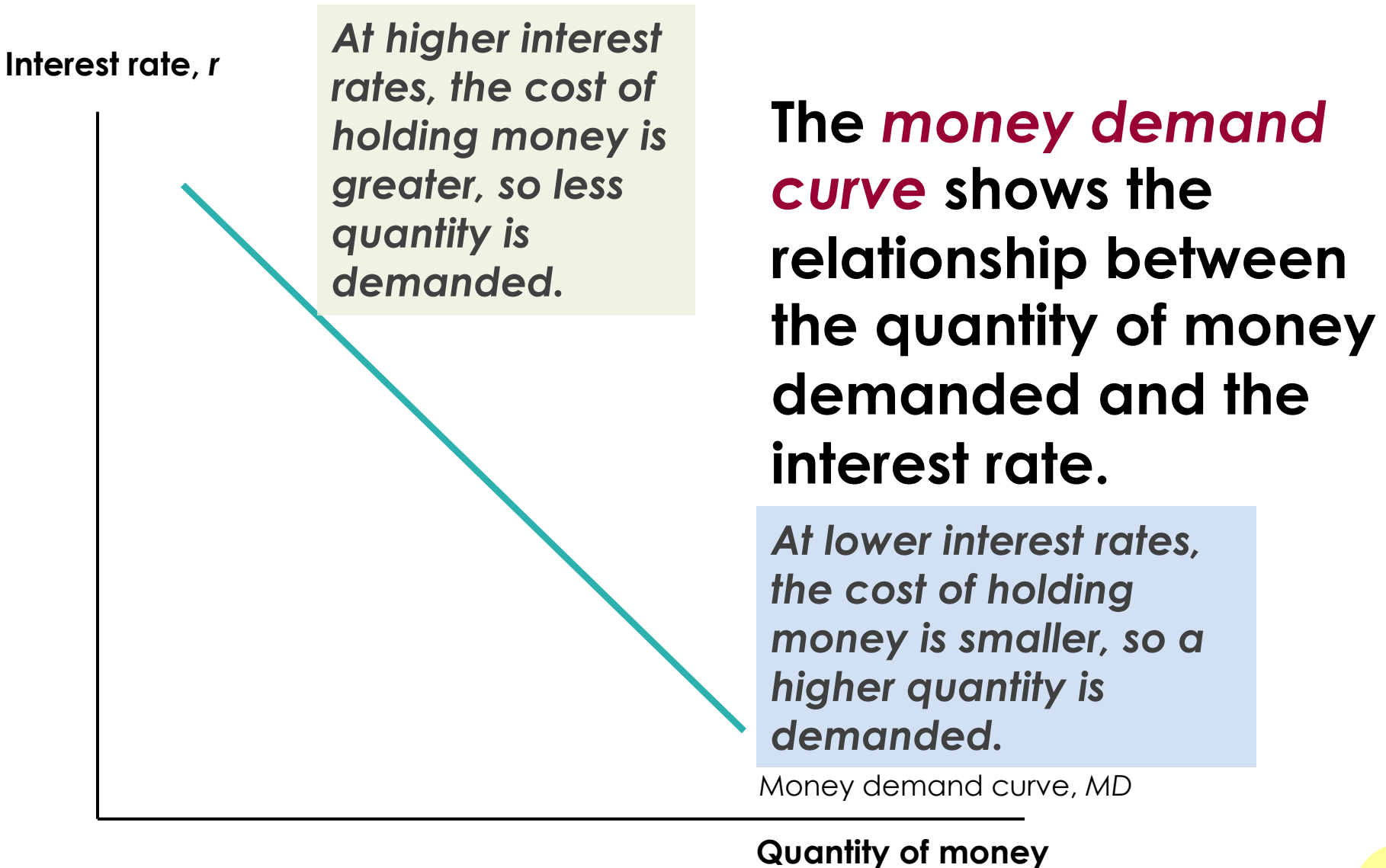
	June 2007	June 2008
Federal funds rate	5.25%	2.00%
One-month certificates of deposit (CDs)	5.30%	2.50%
Interest-bearing demand deposits	2.30%	1.24%
Currency	0	0
CDs minus interest-bearing demand deposits (percentage points)	3.00	1.26
CDs minus currency (percentage points)	5.30	2.50

Source: Federal Reserve Bank of St. Louis

C O P Y R I G H T 2 0 1 5 W O R T H P U B L I S H E R S

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THE MONEY DEMAND CURVE



SHIFTS OF THE REAL MONEY DEMAND CURVE

The following factors shift the money demand curve:

Changes in aggregate price level

Higher prices mean we need more money for transactions (and vice versa).

Changes in real GDP

More goods and services produced and sold means we need more money (and vice versa).

Changes in technology

The ease of credit card payments reduces the need for cash.

Changes in institutions

After 1980 banks were allowed to offer interest on checking accounts. This decreased the cost of holding money, and money demand increased.



ECONOMICS IN ACTION

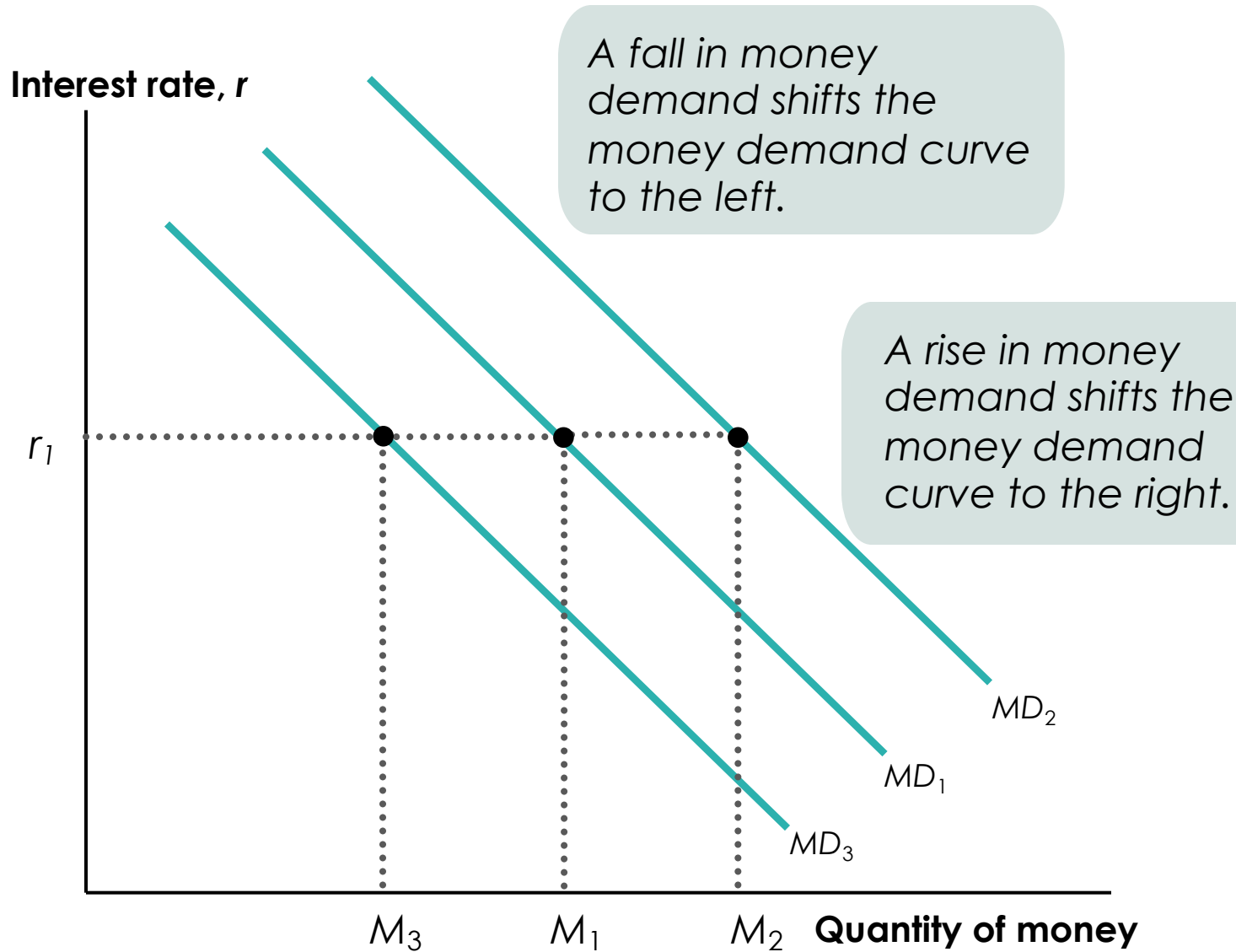
A YEN FOR CASH- Japan is still a cash society.

The many small businesses haven't invested in credit card technology.

And the short-term interest rates have been below 1% since the 1990s. Also, crime is low.

So...why not hold cash?

SHIFTS IN THE DEMAND FOR MONEY



LEARN BY DOING: DISCUSS

With a partner, fill in the table.

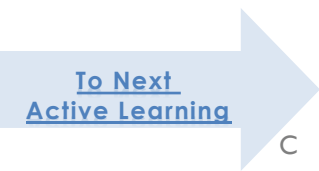
Situation	Shift in the demand for money (provide direction of shift)	Shift in the supply of money (provide direction of shift)	Effect on the equilibrium interest rate	Effect on the equilibrium quantity of money
The information technology for making transactions improves				
The Federal Reserve makes an open-market purchase of Treasury bills				
Real GDP increases in the economy				
Real GDP decreases in the economy, and the Federal Reserve makes an open-market purchase of Treasury bills				
The aggregate price level increases				

[To answer](#)

LEARN BY

(Answers to previous slide)

Situation	Shift in the demand for money (provide direction of shift)	Shift in the supply of money (provide direction of shift)	Effect on the equilibrium interest rate	Effect on the equilibrium quantity of money
The information technology for making transactions improves	Demand for money shifts left, since at every interest rate people will wish to hold less money	There is a movement along the supply of money curve due to the shift in the demand for money	The equilibrium interest rate will decrease	The equilibrium quantity of money will be unchanged
The Federal Reserve makes an open-market purchase of Treasury bills	There is a movement along the demand for money curve due to the shift in the supply of money	Supply of money will shift to the right, as the Federal Reserve injects new reserves into the banking system	The equilibrium interest rate will decrease	The equilibrium quantity of money will increase
Real GDP increases in the economy	Demand for money will shift to the right when real GDP increases	There is a movement along the supply of money curve due to the shift in the demand for money	The equilibrium interest rate will increase	The equilibrium quantity of money will be unchanged
Real GDP decreases in the economy, and the Federal Reserve makes an open-market purchase of Treasury bills	Demand for money will shift to the left when real GDP decreases	Supply of money will shift to the right when the Federal Reserve makes an open-market purchase	The equilibrium interest rate will decrease	The equilibrium quantity of money will increase
The aggregate price level increases	Demand for money shifts to the right when the aggregate price level increases	There will be a movement along the supply of money curve due to the shift in the demand	The equilibrium interest rate will increase	The equilibrium quantity of money will be unchanged



MONEY AND INTEREST RATES

So how does the Fed control interest rates?

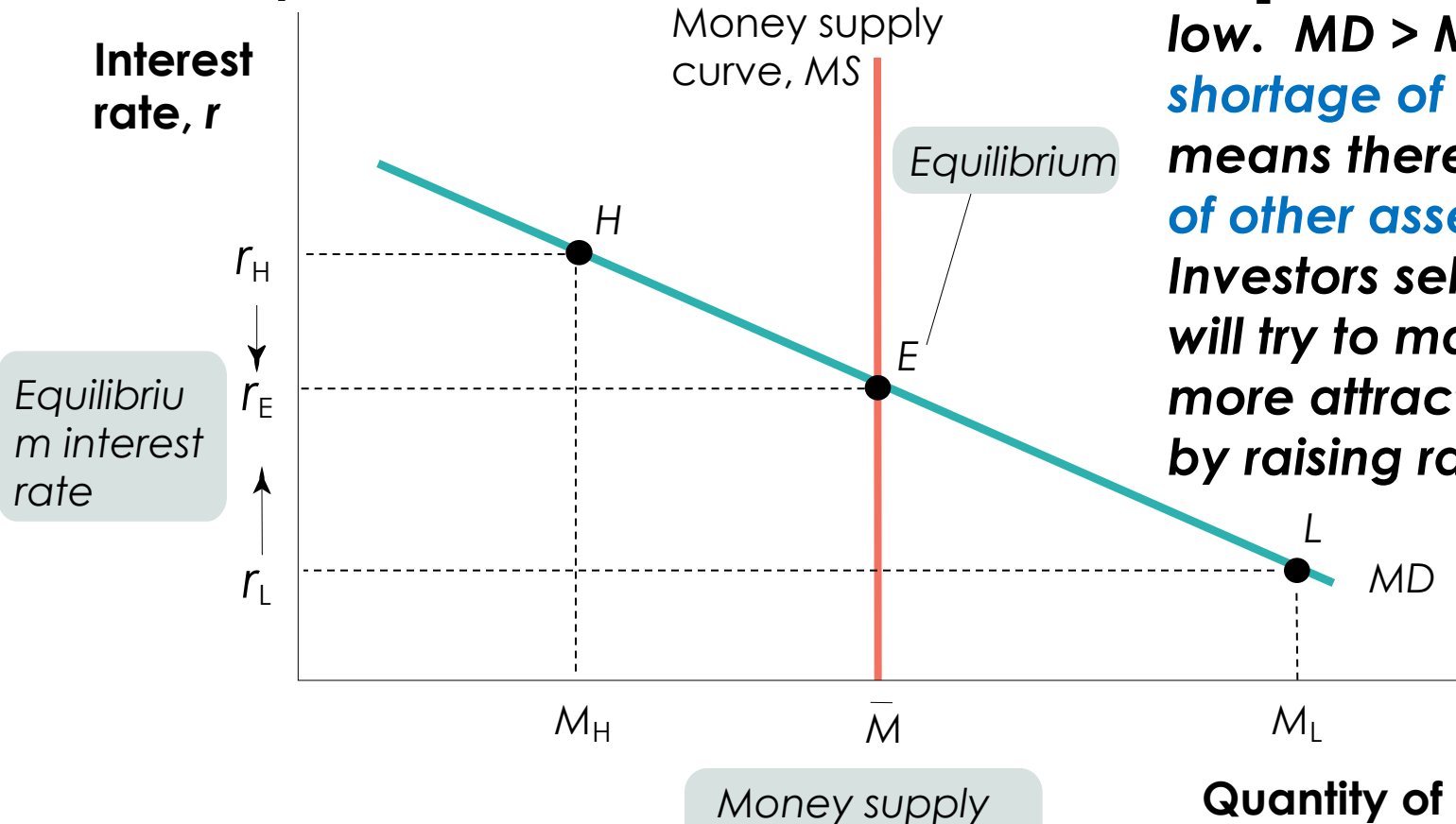
We need to understand how interest rates are set in the market.

The liquidity preference model of the interest rate asserts that the interest rate is determined by the supply and demand for money.

The money supply curve shows how the nominal quantity of money supplied varies with the interest rate.

EQUILIBRIUM IN THE MONEY MARKET

At r_H , interest rates are high. $MS > MD$, and a **surplus of money** means there is a **shortage of other assets** like bonds. Investors selling bonds will realize they can reduce interest rates and still find buyers.



At r_L , interest rates are low. $MD > MS$, and a **shortage of money** means there is a **surplus of other assets** like bonds. Investors selling bonds will try to make them more attractive to buyers by raising rates.

LEARN BY DOING: PRACTICE QUESTION

Suppose that an economy's aggregate output level increases. Assuming that the central bank does not engage in monetary policy, which of the following statements is true?

- a) The interest rate will increase.**
- b) The interest rate will decrease.**
- c) The interest rate will be unaffected by this change in aggregate output.**
- d) This change in aggregate output will cause a movement along the money demand curve.**

MONETARY POLICY AND THE INTEREST RATE

By adjusting the money supply up or down, the Fed can set the interest rate.

The Fed sets a **target federal funds rate** and **undertakes the appropriate open-market operations**.

Target federal funds rate: the Federal Reserve's desired federal funds rate.

MONETARY POLICY AND THE INTEREST RATE

The Effect of an Increase in the Money Supply on the Interest Rate

Interest
rate, r

An increase
in the money
supply ...

MS_1

MS_2

r_1

E_1

r_2

E_2

MD

$\bar{M}_1$

$\bar{M}_2$

Quantity of money

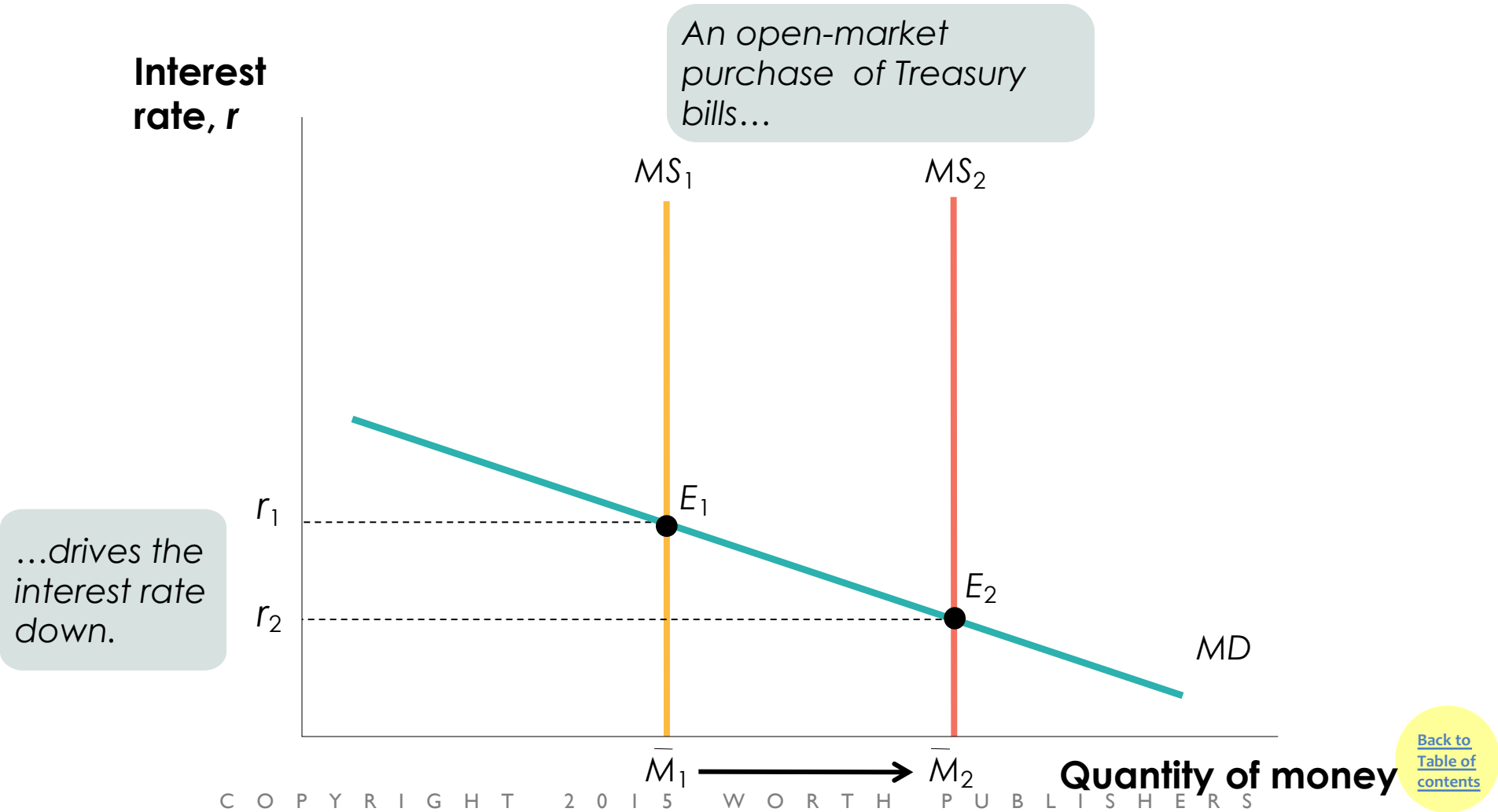
C O P Y R I G H T 2 0 1 5 W O R T H P U B L I S H E R S

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...leads to
a fall in the
Interest
rate.

SETTING THE FEDERAL FUNDS RATE

The Fed uses open-market operations to push the interest rate down to the target rate.

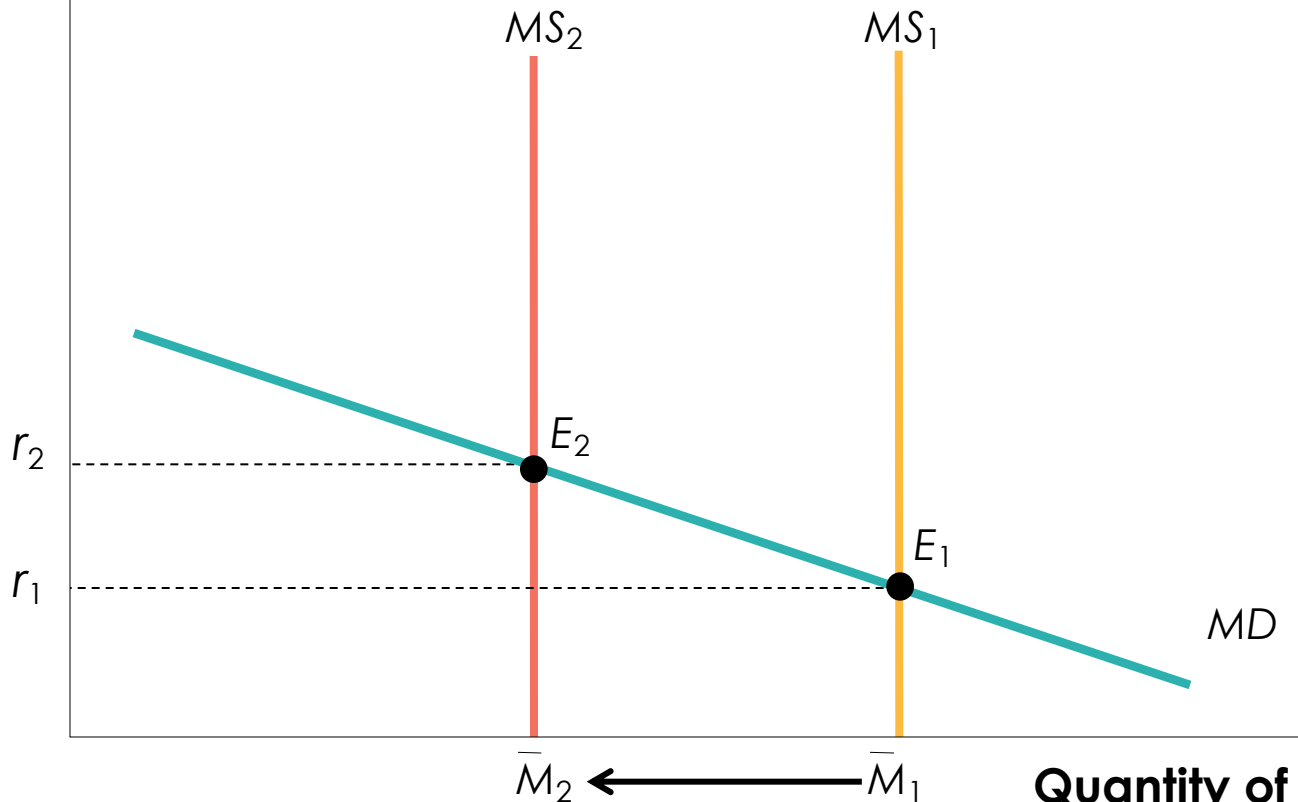


SETTING THE FEDERAL FUNDS RATE

The Fed uses open-market operations to pull the interest rate up to the target rate.

Interest rate, r

An open-market sale of Treasury bills...



...drives the interest rate up.

THE TARGET VERSUS THE MARKET

A common mistake is to imagine that changes in the way the Federal Reserve operates alter the way the money market works.

In fact, the money market works the same way as always: The interest rate is determined by the supply and demand for money.

The only difference is that now the Fed adjusts the supply of money to achieve its target interest rate.

LEARN BY DOING: PRACTICE QUESTION

The money supply curve shifts to the left. The most likely cause for this shift is the FOMC's open-market:

- a) purchase of Treasury bills.**
- b) sale of Treasury bills.**

LONG-TERM INTEREST RATES

Long-term interest rates don't necessarily move with short-term interest rates.

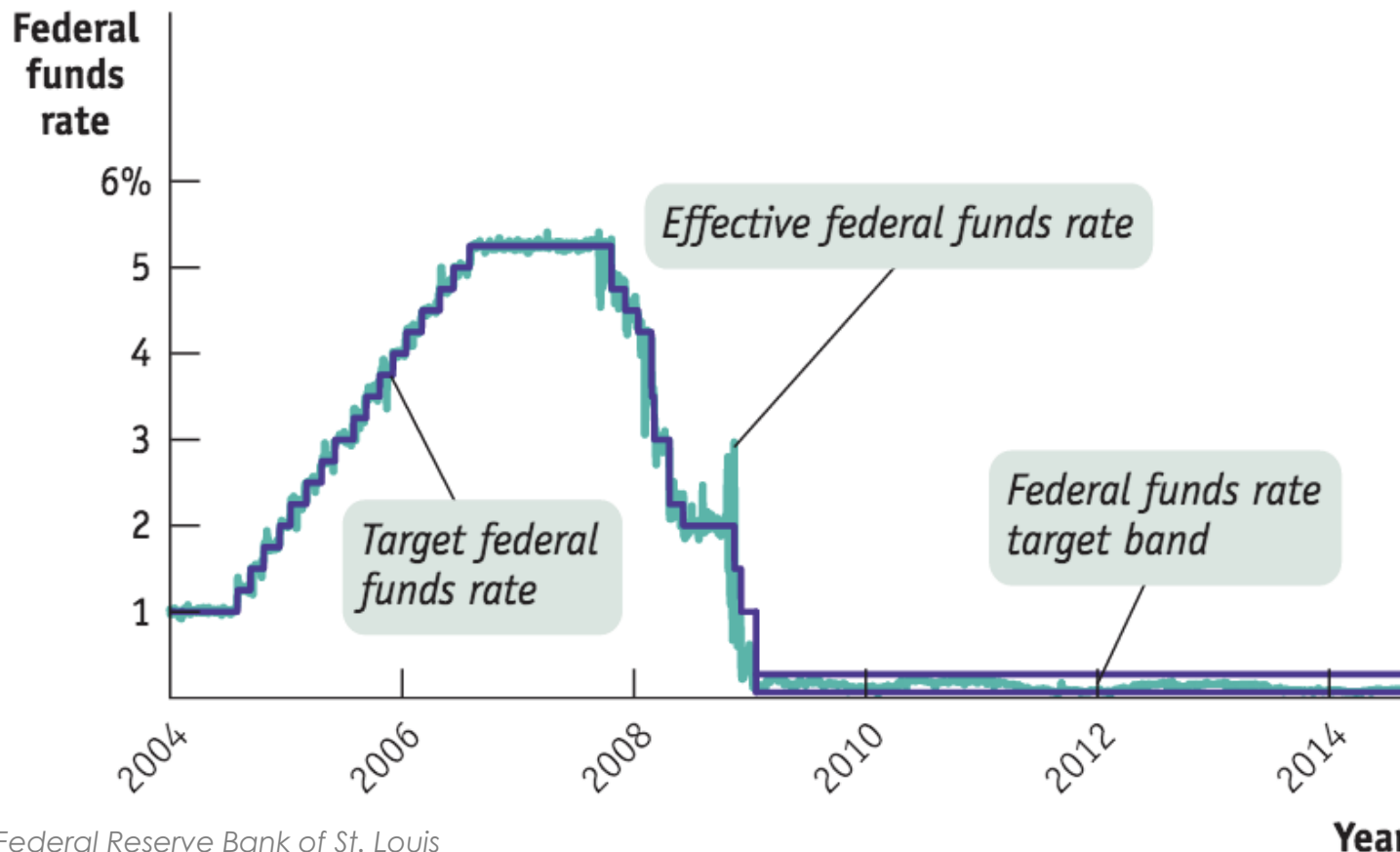
If investors expect short-term interest rates to rise, they may buy short-term bonds.

In practice, long-term interest rates reflect the market's average expectation for short-term rates.



THE FED REVERSES COURSE

Starting in September 2007, fighting the financial crisis took priority.



Source: Federal Reserve Bank of St. Louis

MONETARY POLICY AND AGGREGATE DEMAND

Expansionary monetary policy: monetary policy that increases aggregate demand (also called “easy money policy”).



Contractionary monetary policy: monetary policy that reduces aggregate demand (also called “tight money policy”).

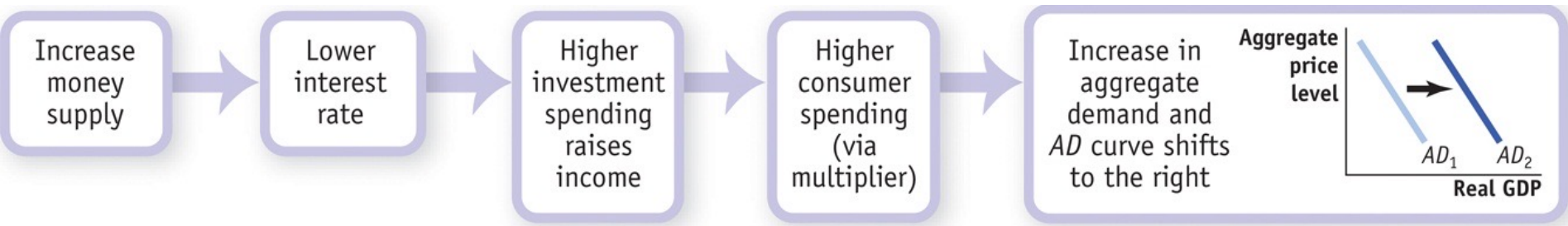
LEARN BY DOING: PRACTICE QUESTION

When the economy is producing at a short-run level of aggregate output that is less than potential output, it is most likely that the Fed will engage in open-market:

- a) purchases of Treasury bills.**
- b) sales of Treasury bills.**

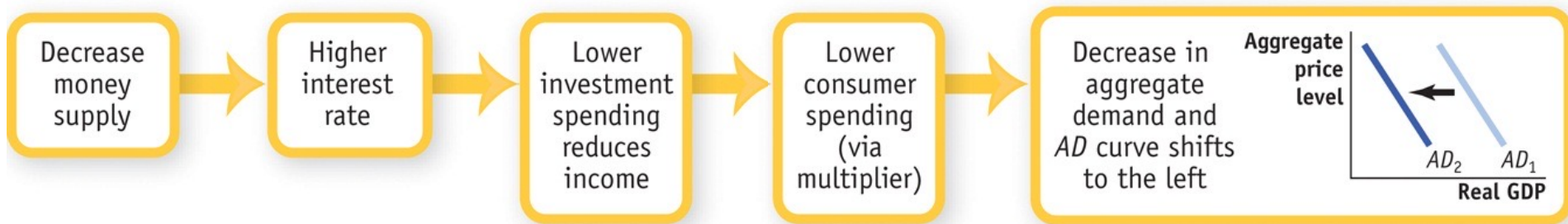
MONETARY POLICY AND AGGREGATE DEMAND

How the expansionary monetary policy cycle works:



MONETARY POLICY AND AGGREGATE DEMAND

How the contractionary monetary policy cycle works:



MONETARY POLICY IN PRACTICE

How does the Fed decide whether to use expansionary or contractionary monetary policy?

And how does it decide how much is enough?

Policy makers try to fight recessions, as well as to ensure *price stability*: low (though usually not zero) inflation.

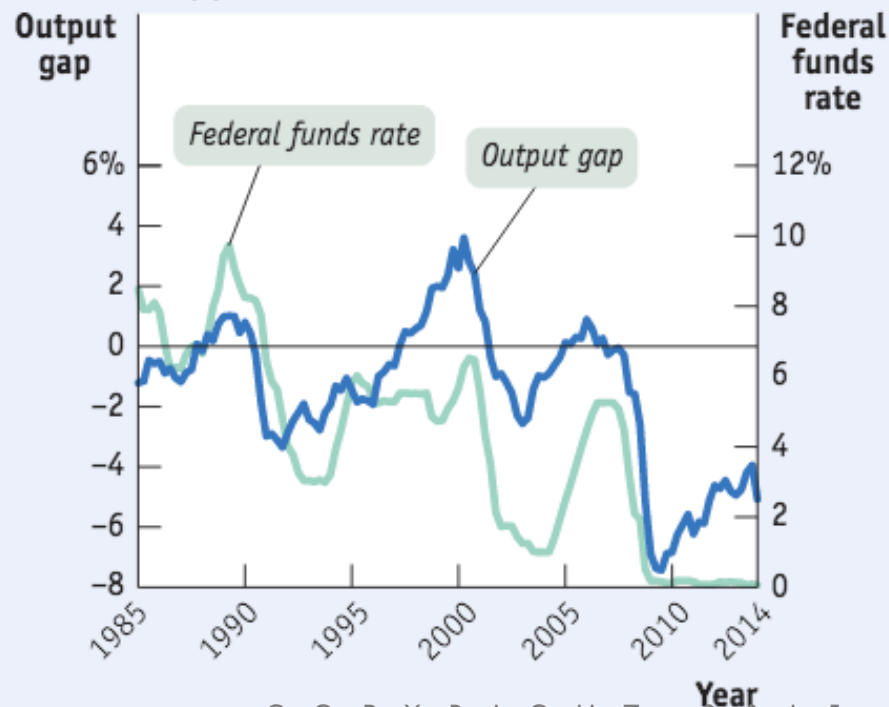
Actual monetary policy reflects a combination of these goals.

TRACKING MONETARY POLICY

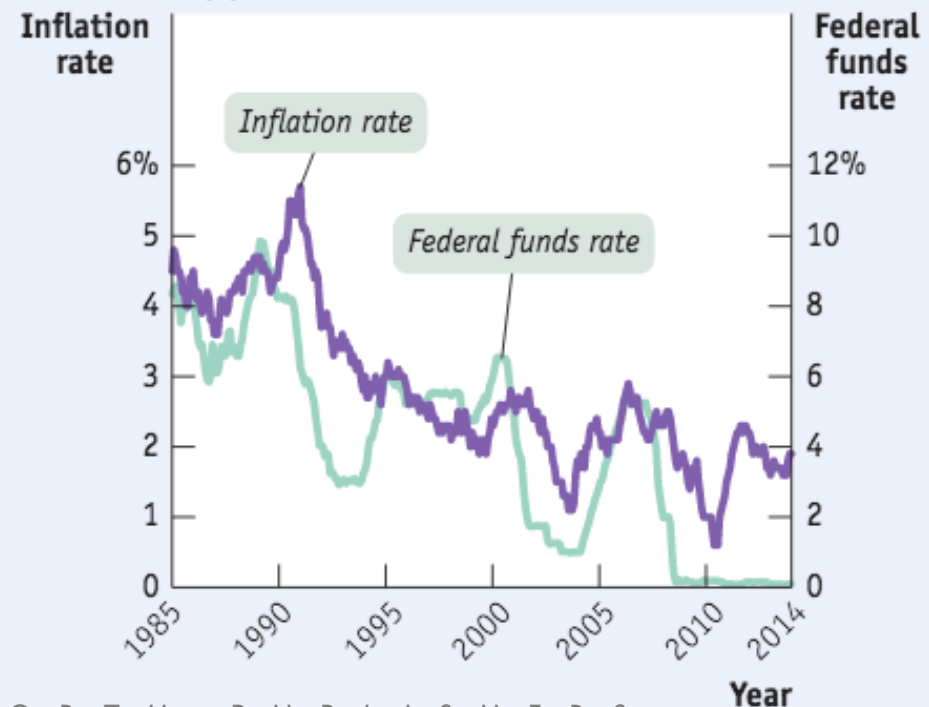
The federal funds rate usually **rises when the output gap is positive** and **falls when the output gap is negative**.

The federal funds rate tends to be **high when inflation is high** and **low when inflation is low**.

(a) Output Gap vs. Federal Funds Rate



(b) Inflation Rate vs. Federal Funds Rate



THE TAYLOR RULE FOR MONETARY POLICY

In 1993, Stanford economist John Taylor proposed a simple rule for monetary policy.

Taylor rule for monetary policy: set the federal funds rate according to the level of the inflation rate and either the output gap or the unemployment rate.

THE FEDERAL FUNDS TARGET AND THE TAYLOR RULE

To address both goals:

- inflation
- recessions

The Fed can follow a rule:



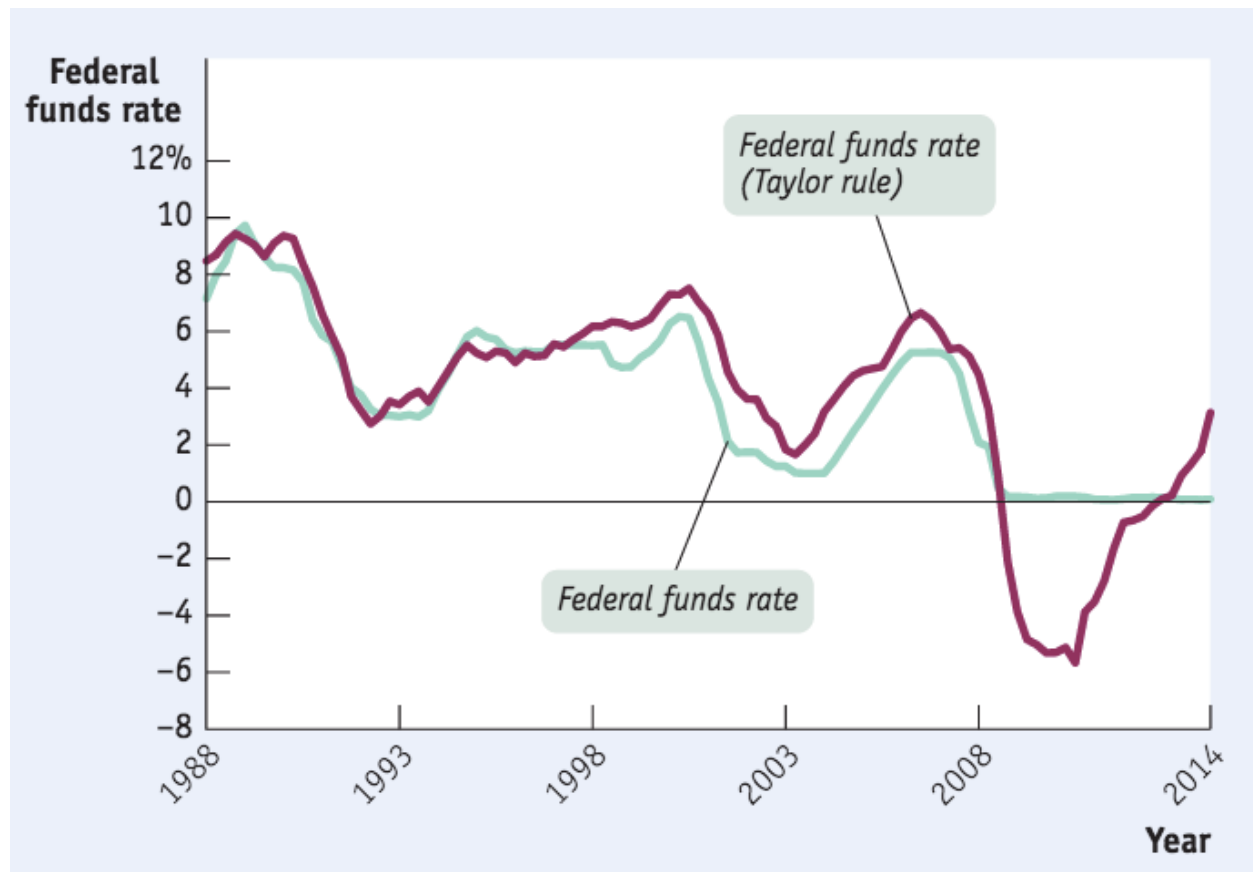
*Federal funds target rate =
 $2.07 + 1.28 \times \text{inflation rate} - 1.95 \times \text{unemployment gap}.$*

As it turns out, between 1988 and 2008 the Fed basically did just this.

TRACKING MONETARY POLICY

The actual federal funds rate tracked the predicted rate quite closely through the end of 2008.

After that the Taylor rule called for negative interest rates, which aren't possible.



Sources: Bureau of Labor Statistics; Congressional Budget Office; Federal Reserve Bank of St. Louis; Glenn D. Rudebusch, "The Fed's Monetary Policy Response to the Current Crisis," FRBSF Economic Letter #2009-17 (May 22, 2009).

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LEARN BY DOING: DISCUSS

With a partner, draw the *SRAS*, *AD*, and *LRAS* curves for an economy with a recessionary gap. What monetary policy should the Fed pursue with its open market operations?

Explain what happens to interest rates. Show on a money demand–money supply graph.

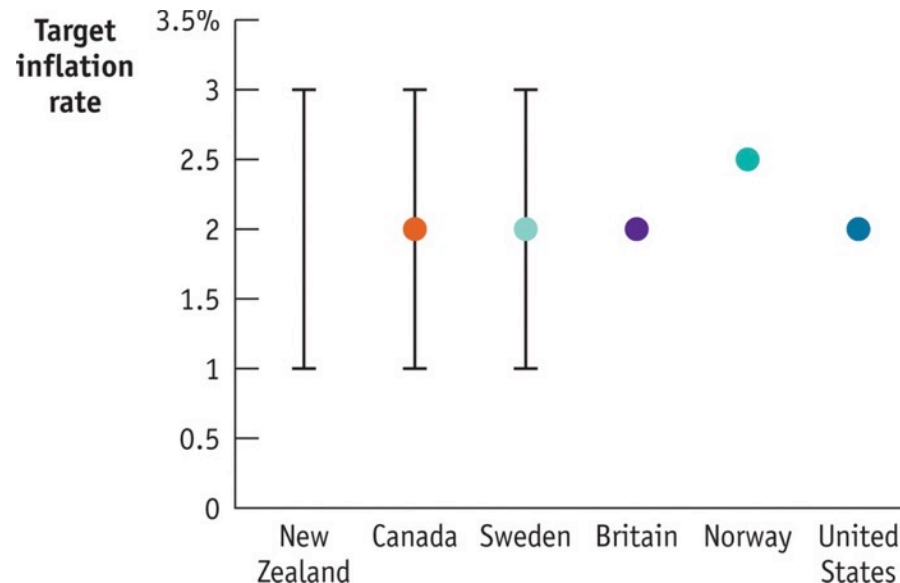
Finally, show the effect of the monetary policy on the original graph.

INFLATION TARGETING



Instead of using a Taylor rule, many central banks use inflation targeting.

Inflation targeting: the central bank setting an **explicit target for the inflation rate** and using monetary policy to hit that target.



DOES IT MATTER?

One major difference between the two methods:

- **Inflation targeting is forward-looking** (based on a forecast of future inflation).
- **Taylor rule is backward-looking** (adjusts monetary policy in response to *past* inflation).

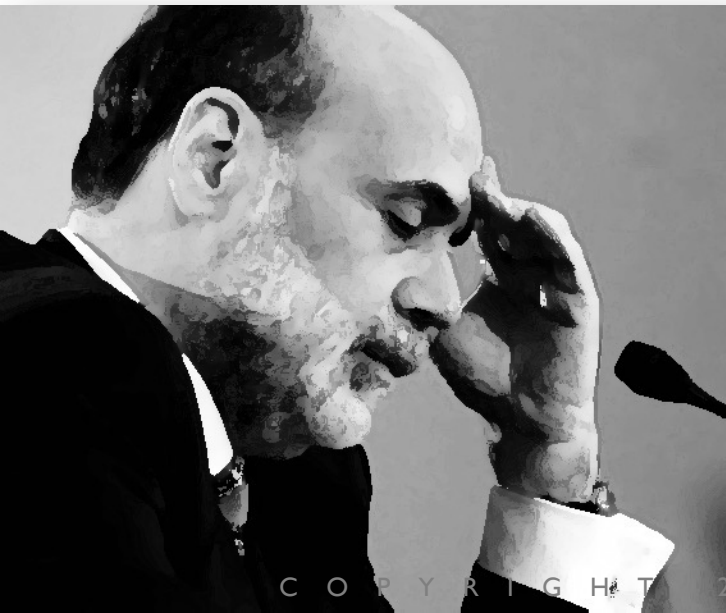


THE ZERO LOWER BOUND PROBLEM

The fact that interest rates can't go below zero (“**zero lower bound**”) limits the power of monetary policy.

November 2010: Interest rates were at or near zero, but the economy was still far below potential output.

So the Fed got creative and tried “quantitative easing” (buying longer-term government bonds).

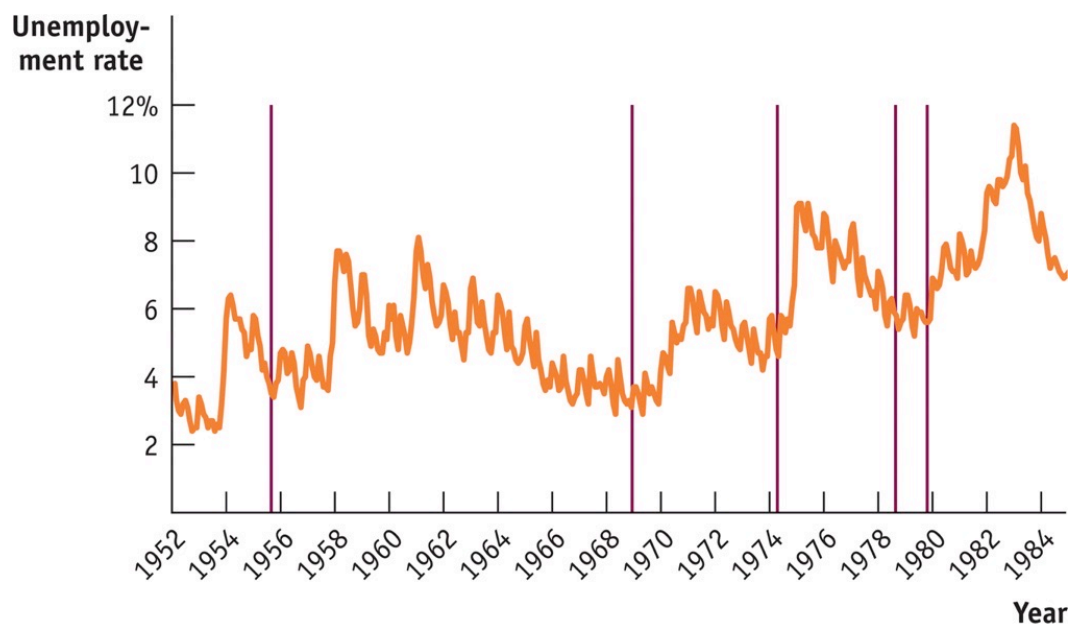


The Fed hoped this would drive down long-term interest rates and have an expansionary effect on the economy.



WHAT THE FED WANTS, THE FED GETS

Causality problems make it hard to observe whether high interest rates are a result of an expansion or of the Fed. Romer and Romer focused on episodes of high interest rates when the **Fed announced it would reduce inflation by increasing unemployment (red lines)**. In short, the Fed gets what it wants.



Sources: Bureau of Labor Statistics; Christina D. Romer and David H. Romer, "Monetary Policy Matters," *Journal of Monetary Economics* 34 (August 1994): 75–88.

MONEY, OUTPUT, AND PRICES IN THE LONG RUN

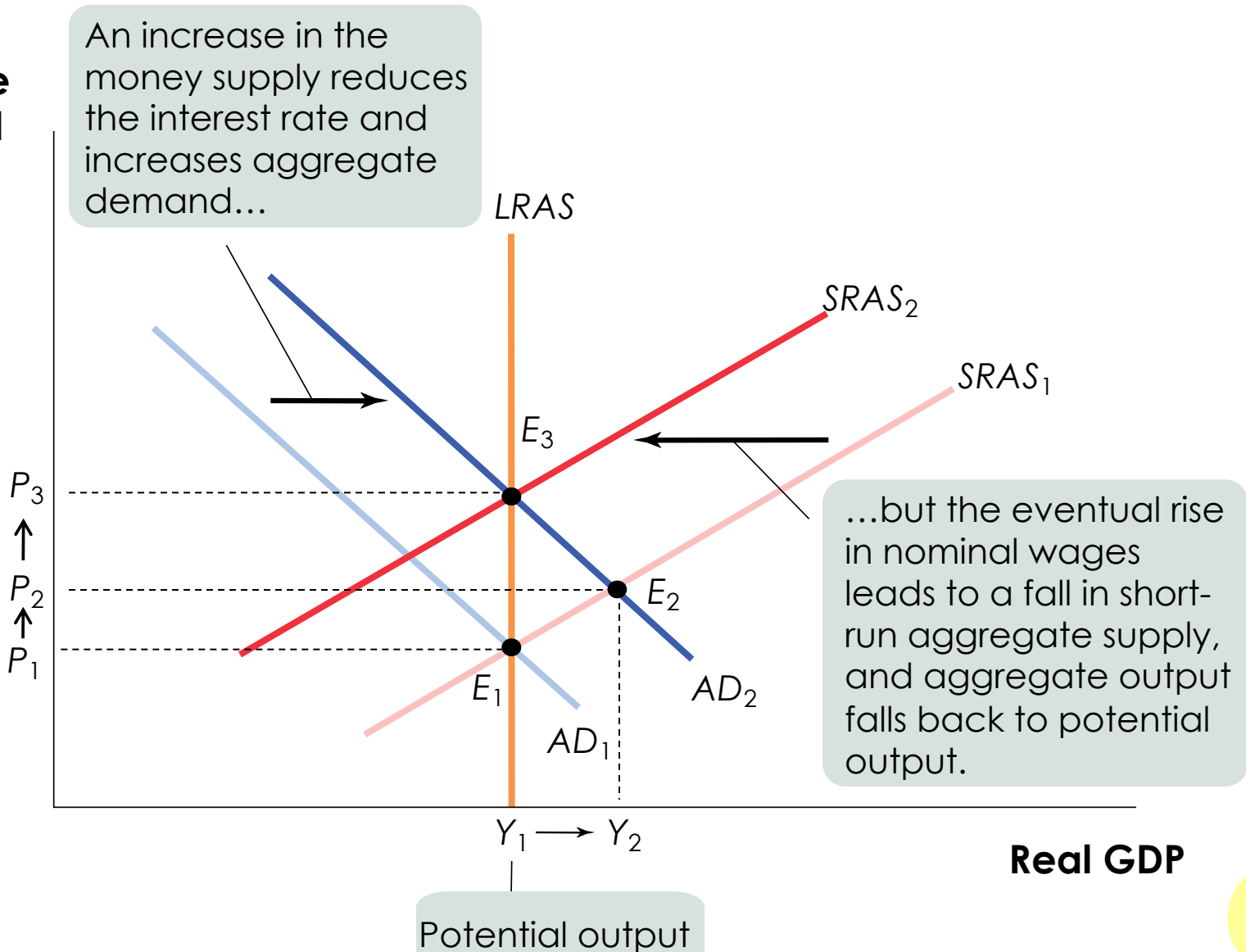
What about the economy's ability to self-correct toward long-run equilibrium through flexible wages and *SRAS* shifts?

In short, we believe that in the long run, changes in the quantity of money affect the price level but not the output or interest rate.



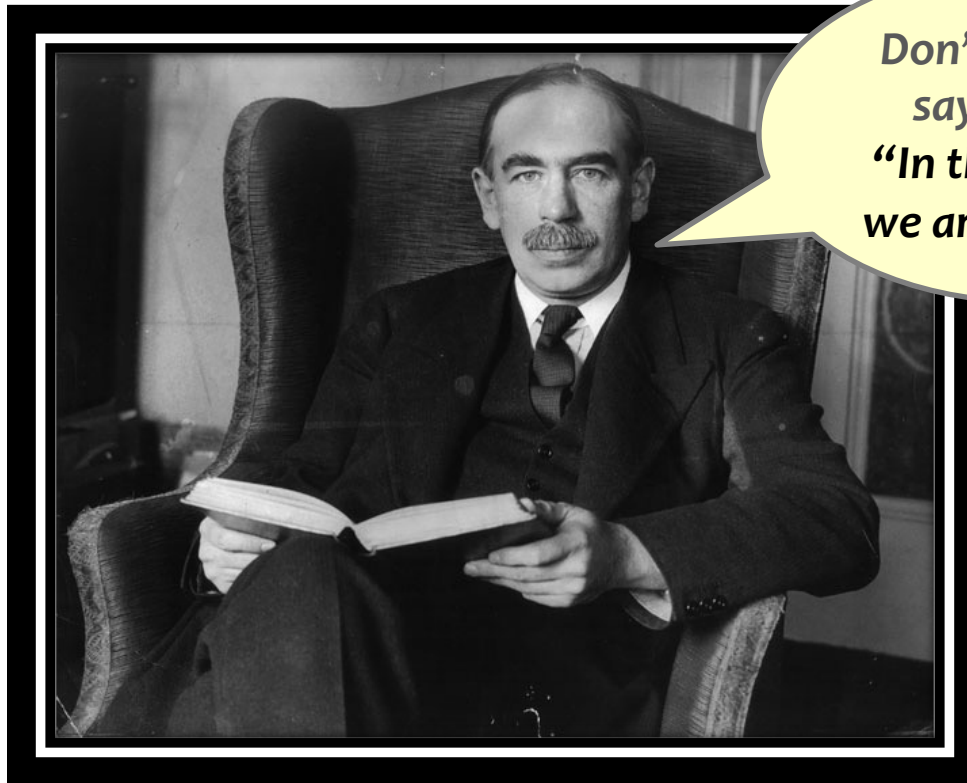
MONEY, OUTPUT, AND PRICES IN THE LONG RUN

Aggregate price level



MONETARY NEUTRALITY

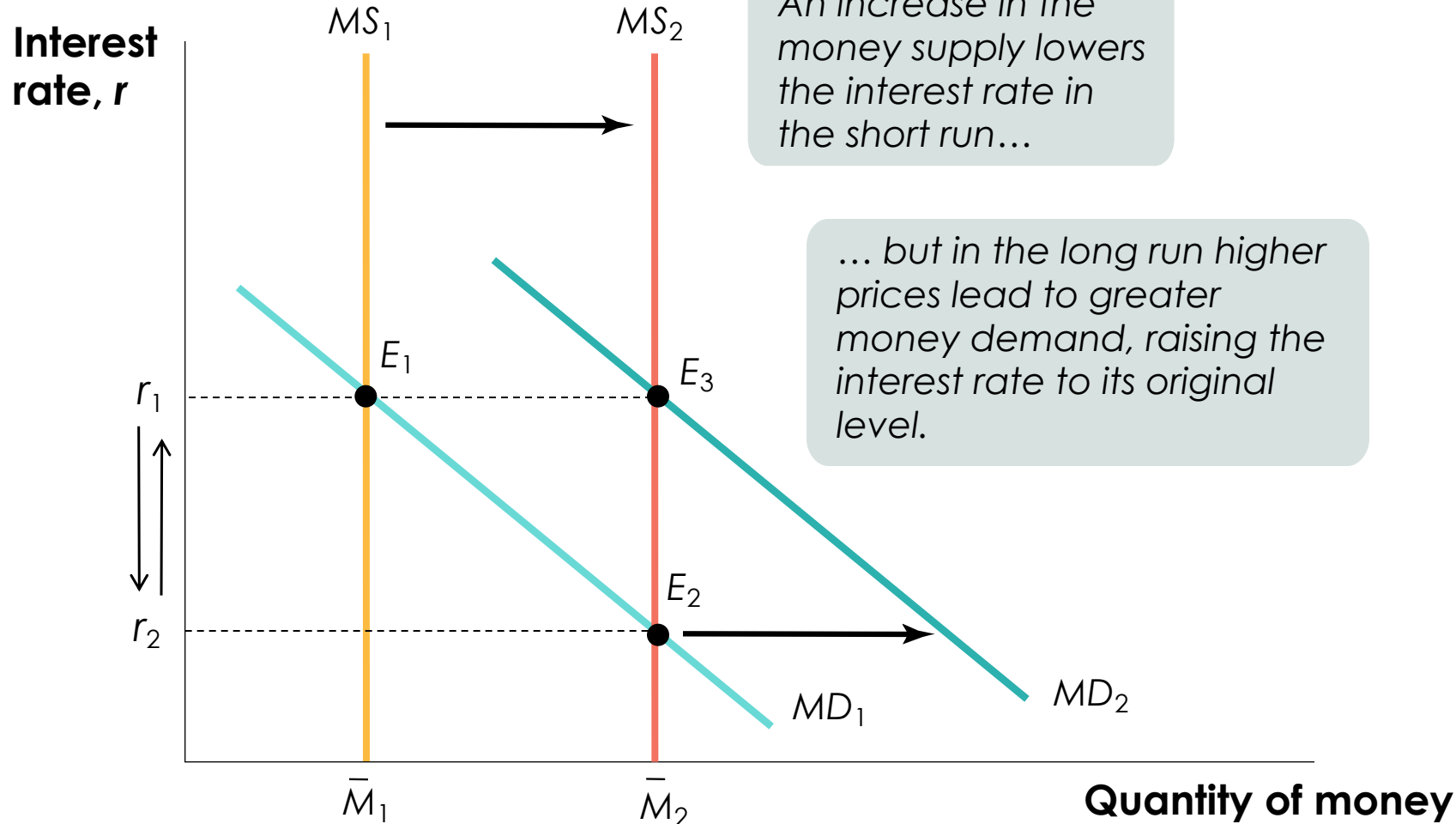
Monetary neutrality: Changes in the money supply have no real effect on the economy.



Don't make me
say it again:
"In the long run
we are all dead."

LONG-RUN CHANGES IN THE MONEY SUPPLY AND INTEREST RATE

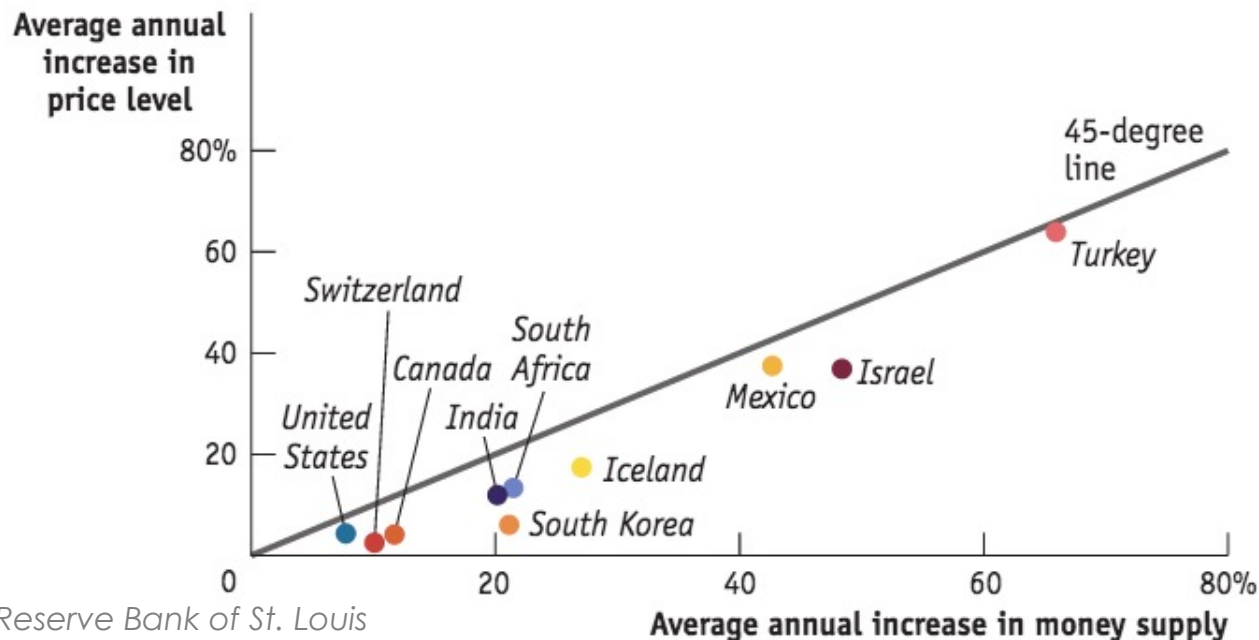
In the long run, changes in the money supply don't affect the interest rate.



ECONOMICS IN ACTION

INTERNATIONAL EVIDENCE OF MONETARY NEUTRALITY

Do increases in the money supply really lead to increases in the price level? It's not exact, but yes.



Source: Federal Reserve Bank of St. Louis

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PIMCO Bets on Cheap Money

In Fall 2009, Bill Gross (then head of Pacific Investment Management Company) believed interest rates would continue to fall- and invested heavily in long-term government bonds.

Meanwhile, Morgan Stanley bet the opposite- and was wrong (until 2011)

